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A Smart Face Biometric Verification Oriented Attendance Handling System Using Artificial Intelligence

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Abstract-The use of facial recognition for biometric authentication in applications like access control and attendance tracking has made it a popular area of study in recent years. Even though monitoring a simple attendance record might be tedious and time-consuming, attendance management solutions are crucial to the success of any business. Biometrics, radio frequency identification, eye tracking, and voice recognition are just few of the numerous automated methods used to identify people. When it comes to verifying a person's identification, facial recognition is among the most used methods. The system underlying this will be advanced facial recognition technology. One of a person's distinguishing characteristics is their facial features. Since there is a little chance of face deviating or being replicated, it may be used to track individuals. In this system, we provide a unique AI-based technique called Artificial Intelligence based Face Biometric Verification (AIFBV), which is responsible for building the face databases that will be used in the subsequent data analysis performed by the suggested recognizer algorithm. Then, during the session when attendance is being taken, photographs of attendees' faces will be matched to those in the database. When a person is recognized, their attendance will be recorded and exported to an Excel spreadsheet mechanically. The daily attendance tally is recorded in an Excel spreadsheet and sent to the appropriate teachers at the close of class. Face detection and recognition using the identified face are the key implementation phases utilized in this sort of system, and dlib is used for both. After these steps are completed, a comparison of the identified faces to a database of student photos should reveal any hidden connections. This strategy will prove to be an effective method for monitoring student attendance.

Index Terms—Artificial Intelligence, AI, Face Biometric Verification, AIFBV, Smart Biometric System, Face Detection, User Verification, Identity Analysis.

I.INTRODUCTION

A manual or automated method for keeping track of employees' whereabouts is essential for any business. For the purposes of assessment and quality control [1], regular class attendance is mandatory for all students. Traditional procedures, such as calling out names or signing papers, are utilized by most companies, despite being inefficient and potentially unsafe. Automatic human identification systems, on the other hand, typically rely on more archaic means like finger prints, passwords, and ID scans. However, there are certain drawbacks to each of these approaches. For example, you may lose your ID card or forget your password. Therefore, a sophisticated facial recognition system [2] is the best option to guarantee complete security and to save records of the past. As an extremely accurate method to identify and authenticate persons, it has played an essential role in security and has been expanding fast in recent years.

Marking attendance manually is a time-consuming process that is still used at many educational institutions today. The teachers also have more work to do since they have to take attendance by calling out students' names, which can take up to five minutes of class time. It takes a long time to do this. There is a possibility that a proxy will show up instead. That's why plenty of schools have started using fancy new methods, like RFID tags, iris scans, fingerprint readers, and so on, to keep track of students' whereabouts throughout class. However, those systems are typically queue based, which may be time consuming and obtrusive.

The most common daily use for a human face is in the context of personal identification. One kind of biometric identification is face recognition, which works by memorizing an individual's unique facial characteristics in the form of a "face print." The widespread potential of biometric facial recognition technology has piqued the interest of many academics. The non-contact nature of face recognition technology makes it superior to other biometric oriented recognition techniques such as fingerprint, palm print, and iris scan.

Recognition methods based on facial analysis may also identify a person at a distance, without the need for direct human interaction. Social networking platforms like FaceBook, as well as public transportation hubs like airports and train stations, are currently making use of facial recognition technology.

Facial recognition has established itself as a crucial biometric feature, one that is both accessible and unobtrusive. Systems that rely on face recognition are not very sensitive to subtle changes in expression. The two main components of any facial recognition system are authentication and subject recognition. The verification of faces is a one-to-one procedure in which a query face picture is matched against a database of stored face templates, whereas is a many-to-many (N:M) challenge.

The goal for this method is to create an attendance tracking mechanism that makes use of facial recognition technology. Here, a person's appearance will be taken into account when recording attendance [5]. In recent years, facial recognition technology has seen increasing adoption and utilization. In this research, we propose an approach called Artificial Intelligence based Face Biometric Verification (AIFBV), which uses facial recognition software to check if a person's face is in a database and then marks them as present in a real-time classroom feed. When compared to conventional procedures, this innovative technology will save time.

II. RELATED STUDY

An analysis of human traits and attributes is called a biometric. Face recognition is a developing area of biometrics for security, despite the reality that no face can be prevented for a security measure [6]. Attendance systems are essential in schools and colleges since manual systems have many disadvantages, including being less accurate as well as more difficult to maintain. Therefore, a variety of systems, including Internet of Things and PIR detector bases, in addition to different models, are present in modern times. In order to prevent damage to the sensor, we need to keep it in a good shape [6]. We face challenges in various models, such deciding which feature to employ ormorecruciallyhandling variation in light, postures, and sizes. We are therefore working to develop an "In-Class" solution to deal with the previously mentioned problem and offer a legitimate digital attendance sheet [6].

The use of facial recognition (FR) for identity is growing in popularity. In actuality, surveillance tasks based on facial recognition can be constructed utilizing the FR method. In this study [7], we present an integrated method that feeds data into a dedicated server, shows real-time information on Android mobile devices, and acquires data from edge devices. As a biometric method for smart attendance, we also use facial recognition. We proposed applying convolutional neural network-based deep learning techniques to this system. Faces are taken via CCTV data streaming and compared to the database. As a result, it is regarded as their attendance log. It is also annotated and entered into the database. Big data technology has built this system prototype to address this data complexity. Real-time monitoring is possible for the faces that have been identified [7]. Finally, before reports in real time are provided via the web as well as an Android device through API, hash encryption encrypts data transmission [7].

The development of biometric recognition technology for use in a variety of fields is made possible by its application in personal authentication [8]. Biometric systems for identification can be implemented using physical or behavioral traits including voice, fingerprints, faces, and irises [8] [11]. There's a great opportunity to do interesting research in this area because the education sector currently uses very little biometric recognition-based attendance monitoring equipment. As seen in a typical classroom, teachers usually call names or sign an attendance sheet as traditional methods of keeping track of students' attendance. On the other hand, fraud has been demonstrated to occur sometimes using these methods, and they are time-consuming and difficult. Significant progress has been achieved in the direction of automatically recording attendance by biometric recognition as a consequence. Because of this advancement, a new and more advanced biometric-based attendance system is being developed over the last ten years. Configuring hardware and software is necessary for biometric-based attendance systems. Due to the size of the software and hardware parts being too big for one report, this literature review only provides a broad overview of the many types of hardware being used. Thereafter, attention is placed on the microcontroller, the biometric sensors, the communication channel, database storage, and other components to aid future researchers in developing the hardware part of biometric-based attendance systems [8].

These days, facial recognition and detection are cutting edge technologies. This technology has many applications, including enhancing security, adjusting camera distance, and identifying a person's image in still and moving images [9]. We made the decision to create a device that would detect and identify faces in order to track student attendance. The goal of this work [9] is to create a system that will assist lecturers in taking student attendance in an efficient manner.

The technical landscape is rapidly evolving in a developing nation like India. In light of this, many institutions must automate their attendance systems [10]. When it comes to educational institutions, there are a number of tactics that are employed to manage and track student attendance. The most frequent method of recording attendance is by hand, with some institutions even using finger print recognition software. Attendance systems that use facial recognition are still not common, but they have a lot of potential. Future prospects are greatly enhanced by the Deep Learning research [10], particularly in the field of neural networks arena. Much time and effort can be saved by an intelligent system that combines a powerful facial recognition system with an easy-to-use graphical user interface. A database created with MySQL or an Excel spreadsheet can be used as the backend for such a system, which can be developed in Python utilizing machine learning algorithms. This paper suggests such a system that makes use of the Python Cv2 library of face detection and the De-facto standardized GUI library. After a face is detected to record attendance, a Convolutional Neural Network is constructed with the use of the Keras library to assist in identifying or recognizing a person's facial traits. By using their face traits, the system will identify the user automatically and store their attendance into an Excel spreadsheet. Essentially, this study highlights the progress made in Deep Learning using a real-world application [10].

III. METHODOLOGY

Facial recognition technologies have come a long way in the past few years. In the past decade, facial recognition technology has made great strides forward. Most existing facial recognition algorithms do well even when there are only a handful of faces in view. These techniques have also been tried and proved in lab settings with ideal lighting, appropriate facial expressions, and no blurring. This research proposes a method for attendance systems that uses artificial intelligence to verify the identities of many faces inside a single image, regardless of lighting or placement. The system is named Artificial Intelligence based Face Biometric Verification (AIFBV).

Once all students in the class have enrolled and given the required information, a dataset containing photos of each student will be generated. All throughout each class session, facial recognition will be tested on streaming video from the classroom. We will compare the discovered faces with images from the dataset. Students whose names match will have their attendance noted. At the end of each session, a list of absenteeism will be sent to the lecturers. The following figure Fig.1 shows the architecture of the system. There are four distinct phases to this procedure:

A. Dataset Creation

Webcam footage of the students is recorded. A variety of photos of a single student's motions and perspectives will be captured. These photographs are processed beforehand. In order to use the photos in the identification process, they are cropped to get the Region of Interest. Resizing the clipped photos to a specific width or height in pixels is the next step. After that, these pictures will be changed from RGB to grayscale. The pictures of each pupil will be used as the file names in a folder.

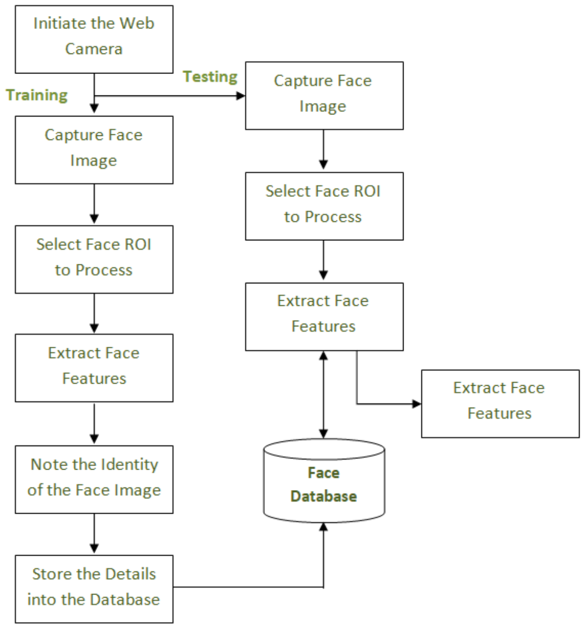

Fig.1. System Architecture

B. Face Detection

The Haar-Cascade Classifier in OpenCV is used for face detection here. In order to be utilized for face detection, the Haar Cascade algorithm must first be taught to recognize human faces. Feature extraction describes this process.

C. Face Recognition

Preparing the data, training the face recognizer, and making predictions are the three stages of face recognition. In this instance, the collection's existing photos will be used as training data. The label will have the student's identifying number on it. After that, a system that recognizes faces uses the images. Local Binary Patterns Histogram is a facial detection algorithm that powers the approach.

D. Attendance Updation

Following the implementation of the facial recognition algorithm, the list of students who were there and those who weren't will be updated on an Excel sheet and provided to the appropriate departments. Faculty will receive a fresh, updated attendance sheet at the end of each month.

In order for the attendance tracking system to function, a collection of data, including the identities and photographs of all employees, must be entered into the system. Taking pictures of people's faces with a camera is the first step in the portrait acquisition process. After that, the photos will go through a series of pre-processing steps to produce a grayscale image as well as cropped faces of uniform size. Following processing, the photos are saved in a hierarchical file structure. In this study, each of the images of people's faces will be kept in a hierarchical structure under a folder called "database." When you expand the database folder, you'll see several subfolders, one for each person in the database; within each of these subfolders, there will be a collection of photos of that person's face. The following figure Fig.2 illustrates the methods for gathering and preprocessing images.

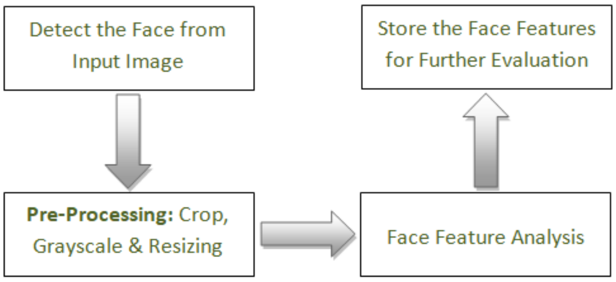

Fig.2 Methods for Gathering and Preprocessing Images

In this method of recording attendance, participants are first enrolled, then their faces are detected and recognized, and finally their attendance is recorded in a database. Each image is therefore transformed into a characteristic in a multidimensional space. In this very detailed region, similar characteristics of the same individual will be near together, while those of different people will be spread far apart.

IV.Results and Discussions

Facial recognition is being utilized in the biometric attendance system prototype that is being developed in Python for Windows.
(i) Recognize individual faces.
(ii) As a method of tallying the results.
(iii) Take a look at the list of students who have been fined.

Recently, digital photography has played a unique role in the identification of technical advances. This is possible as a result of image processing, which draws significant information from the photos themselves. The key objectives of this project are the enhancement of picture information via human translation, as well as the transmission and representation of data obtained via autonomous machine vision. Because of the proliferation of smart phones and compact cameras, more people are using digital cameras. A biometric visitation log for pupils that makes use of technology that recognizes faces has been proposed as the system to implement. Over the course of the last several decades, there has been a significant amount of research conducted on the subject of facial recognition. The purchase of an item has a role in determining how large the candidate's face appears in the picture in this particular case. It is a method for teaching a computer to identify a human face in a photograph by directing the computer to examine several samples of faces that are included in its input data. In this program, accuracy is prioritized by concentrating on reducing the amount of erroneous standards that are used. The objective of artificial intelligence-based face biometric verification, also known as AIFBV, is to reduce the size of a data set that comprises numerous related parameters while maintaining the same degree of variability. The same job can be completed in a sequential fashion that results in a reduced number of maintained variants of the original dynamics if the dynamics in a new set of dynamics, known as key elements and orthogonal, are modified.

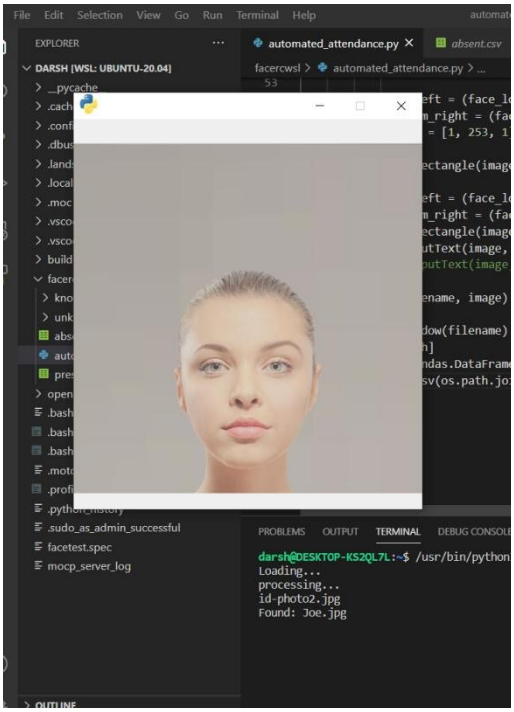

Fig.4 Face Recognition Cam Working Test

This modification of the dynamics allows for the task to be completed. Therefore, the variety from its beginnings may still be found in the very first significant portion. The eigenvectors of the covariance matrix are orthogonal due to the fact that they are representations of the matrix's most significant determinants. Customers are given the opportunity to interact with the system through the usage of graphical user interfaces. This system gives users access to three primary functions: the ability to enroll students and teachers, record attendance and record attendance for students. On the student registration form, the student is responsible for providing each and every requested piece of information. When the "Register" button is hit, the camera starts recording instantly, and the interface seen in Figure 4 appears. And the following figure, Fig-5 illustrates the face recognition process of the proposed approach. It then begins searching the photos for faces and evaluating them accordingly. As soon as 50 samples have been collected, or if CRTL+Q has been pressed, the camera will start capturing photographs on its own. After the preprocessing step, these photographs will be saved in the directory titled "training images" The provided faculty registration form must be completed by each member of the faculty, including the inclusion of their respective course codes and email addresses. It is important to note that the absence list will be distributed to the appropriate departments as a result of this.

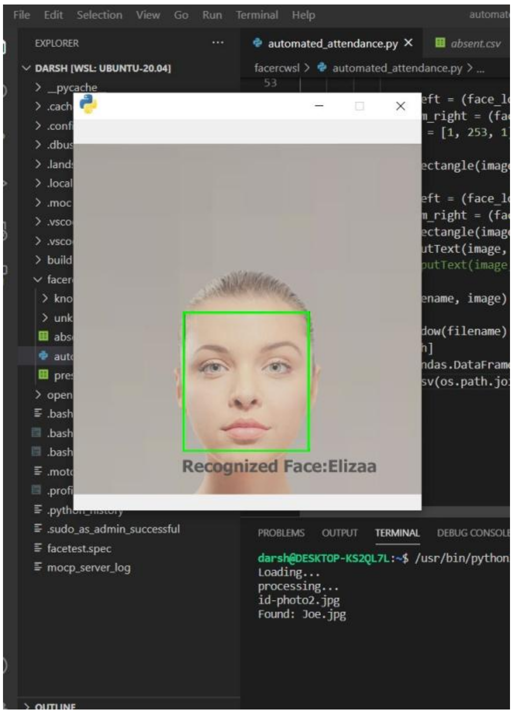

Fig.5 Face Recognition Process

The following figure, Fig-6 illustrates the efficiency of the proposed AI based scheme called AIFBV, in which it will be cross-validated with the conventional learning scheme called CNN to prove the efficiency of the proposed scheme. And the same is illustrated over the following table, Table-1.

Table-1 Face Recognition Efficiency

S.No.	Tested Faces	CNN (%)	AIFBV (%)
1.	25	91.26	96.67
2.	50	91.37	96.54
3.	75	92.71	96.35
4.	100	92.57	95.15
5.	125	92.76	95.45
6.	150	92.64	96.68
7.	175	93.48	96.75
8.	200	93.26	96.89
9.	225	93.39	97.27
10.	250	93.54	97.36

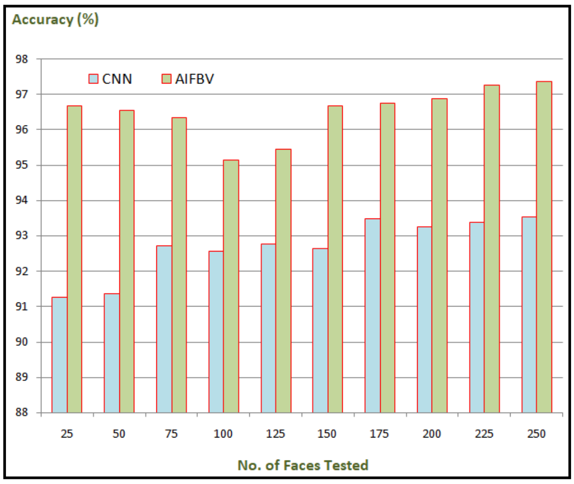

Fig.6 Face Recognition Efficiency

V. Conclusion

In the present research, we suggest a novel Attendance Management System that is powered by Artificial Intelligence (AI) and is based on Face Recognition. We refer to this system as Artificial Intelligence based Face Biometric Verification (AIFBV). Facial recognition now offers the highest level of accuracy compared to other methods, including biometrics and other approaches that are currently accessible; nevertheless, there are many alternative techniques. It is not necessary to make use of any specialist hardware in order to put the system into operation. Only a camera, a laptop, and a database are needed to put together an attendance tracking system that is based on facial recognition.

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